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STANDARD IOTS (STANDA,35) 10/2/73 JGC/JBL

/SWAPPER IOT'S FOR FASTRAND SWAPPING

```
SWAP1:   STF 1           /IN
SWAP3:   JSP ,SETUP      /SWAP
          STF 6
          LAC SWAP4
          DAC I (DCH 71)
          LAC SWAPT
          JDA TRACK
          000000
          JSP WAIT4
          SZF 1
          JMP SWAP2.      /READ FROM UPPER 9 BITS
          LAC I AC
          JDA SWAPR
          260000
          SZF 2
SWAP2.,  JMP SWAPE1
          LAC I AC
          RAR 9S
          JDA SWAPR
          200000
          LAC I (IOR 50) /SET UP INCOMING USER
          DAC I AC
          LAC I (IOR 51)
          DAC I IO
          LAC I (IOR 52)
          DAC I FLAGS
          JMP SWAPE1

SWAP2:   STF 2
          JMP SWAP3"A"7777

SWAPC0,  560001
          0
          0
SWAPC0 3, 040000 .
          100001
          0
SWAPC0 6, 040000 .
          100001
          0
SWAPC0 11, 040000 .
          100001
          000000
```

SWAPR,	0
	DAC SWPRX
	LAC SCORE
	DAC SWAPC0 2
	LAW 777
	AND SWAPR
	MUL C81.
	DIV ONE
C81.,	81.
	IOR I SWPRX
	DAC SWAPC0 1
	AND (-77)
	ADD C100
	DAC SWAPC0+5
	ADD C100
	DAC SWAPC0 10
	LAW 77
	AND SWAPC0 1
	CMA
	ADD C100
	MUL C63
	DIV ONE
C63,	63
	SUB ONE
	DAP SWAPC0 3
	CMA
	ADD (7776)
	DAP SWAPC0 6
	SUB (6277)
	SZM
	JMP SWAPR1
	DZM SWAPC0 10
SWAPR2,	IDX SWPRX
	JSP GO
	I I I SWAPC0
	JMP I SWPRX
SWAPR1,	SUB ONE
	DAP SWAPC0 11
	LAW 6277
	DAP SWAPC0 6
	JMP SWAPR2
SWPRX,	0
SWAPT,	200000

/FIXED HEAD IOT'S

/INTERNAL SUBROUTINE FOR FIXED HEAD IOT'S

```

NUM,      DAP NUMX
          LAC FPAR      /GET ADDRESS NO.
          SUB (2100
          SMA
          JSP ILLSPC
          LAC FPAR
          MUL ONE
          DIV C17.
C17.,     17.
          IOR (217000    /SET TO READ FROM HEAD 7 THIS SECTOR
          DAC RNMC0+1
          LAI           /GET POINTER INTO BLOCK FROM REMAINDER
          SAL 1S
          AAI
          ADD (170000 BUFF
          DAC NUM1      /POINTER TO BLOCK
          DAC NUM2
          JSP GO
          I I I RNMC0
          JSP WAIT4     /WAIT FOR CORE TO SHOW UP
          LAW 7777      /GET LOW 12 BIIS OF AC
          AND I AC
          DAP RPMA      /FIRST WORD
          IDA
          DAP RPMB
          IDA
          DAP RPMC
NUMX,     JMP .
NUM1,     0
NUM2,     0
RNMC0,    560001
          0
          630000 BUFF
          100063
          000000
RPMA,     IOR .
RPMB,     IOR .
RPMC,     IOR .

```

/READ NUMBER
 /IO=ADDRESS NO
 /IF FLAG 2 SET READ AS I OR BLOCK (THEN HAS SAME SEQ AS IOT)

```

RNUM:    JSP ,SETUP
          LAC I (DCH 72) /GET NUMBER
          JDA TRACK
          030000
          JSP NUM          /DECODE AND SET UP FOR READ
          LIO I NUM1       /GET NUMBER
          DIO I IO
          SZF I 3          /DOES HE WANT REWRITE NO.
          JMP RNUM1
          IDX NUM1
          IDX NUM1
          LAC I NUM1
          DAC I AC
RNUM1,   SZF I 2
          JMP SWAPE1       /DONE
          SPI              /WANT TO READ IS IT LEGAL DRUM ADDRESS
          JSP ILLSPC       /NO
          LAW 7777
          AND I AC
          SUB BOUND
          SPA
          JSP ILLSPC       /BELOW BOUND
          LAI"U"CLF 2
          RAL 88
          AND (177
          ADD (FRELST)
          LIF
          SWP
          AND (6)
          SAD (6)
          JSP ILLSPC
          SAL 68
          AAI
          JDA READ
          AND (=177
          DAC THIRD
          JMP RDINF3
  
```

/WRITE FIXED HEAD NUMBER
 /FLAG 1=1,REWRITE,0,WRITE
 /FLAG 1=1;FLAG 2=1,EXPUNGE
 WNUM;

JSP .SETUP
 LAW 7777
 AND I (DCH 70)
 DAC TRACK
 LAC I TRACK
 JDA TRACK

/INDIRECT THRU IT

030000

JSP NUM

/GET BLOCK INTO CORE

LAW I 3

/IS IT LEGAL NAME

DAC COUNT

SZF 3

JMP RDNUM

SZF 1

JMP RWNUM

WNUM2,

CLC

SAS I NUM2

JMP RTERR

IDX NUM2

ISP COUNT

JMP WNUM2

LAC I IO

/0.K.=STORE NUMBER

DAC I NUM1

IDX NUM1

LAC I RPMB

DAC I NUM1

/SAVE OWN=WORD

IDX NUM1

LAC I RPMC

/SAVE REWRITE NUMBER

DAC I NUM1

RWNM4,

LAC (270000

DIP RNMCO+1

/RESET TO WRIT E OUT BLOCK

JSP GO

I I I RNMCO

JMP SWAPE1

RDNUM,

LAC I NUM2

/IOT 6610 READS THREE WRD BLK INTO CORE

DAC I RPMB

IDX NUM2

LAC I NUM2

DAC I RPMB

IDX NUM2

LAC I NUM2

DAC I RPMC

JMP SWAPE1

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```
RWNUM,      IDX NUM2
             LAC I RPMB      /CHECK OWN-WORD
             SAS I NUM2
             JMP RTERR
             IDX NUM2      /LOOK AT REWRITE NUMBER
             LAC I NUM2
             SZA I
             JMP RWNM3
             SAS I RPMC
             JMP RTERR
             IDX I NUM2
             SZA I          /SKIP ZERO
             LAW 1
             DAC I NUM2
             DAC I RPMC
RWNM3,      SZF 2
             JMP ENUM      /EXPUNGE
             LAC I IO
             DAC I NUM1
             JMP RWNM4

/EXPUNGE NUMBER
ENUM,      CLI"U"CM1      /O.K.SET TO -0
           DIO I NUM1
           IDX NUM1
           DIO I NUM1
           IDX NUM1
           DIO I NUM1
           JMP RWNM4      /GO REWRITE THE BLOCK
```

/READ (OR WRITE) A FIXED-HEAD BLOCK. USEFUL FOR DIAGNOSTICS OR
/ FOR "NCP" TO EXPUNGE 256, IVN'S IN A HURRY.

```

FIXHD:   JSP .SETUP
         LAW 7777
         AND I (DCH 70)
         SUB BOUND      /LEGAL CORE ADR ? (NO IOPMAX CHECK)
         SPA
         JMP I 16SPCERR
         LAW I 7077
         AND I (DCH 72) /LEGAL BLOCK NUMBER
         SZA
         JMP I 16SPCERR
         SZF I 1        /WRITING ?
         JMP FIXHD2      /NO, MERELY READING.
         LAC I NETHND    /YES, IS THIS THE "NCP" ?
         SAS 16STAT
         JMP I 16SPCERR
         LAC I (DCH 72) /IS THE "NCP" CHANGING THE NETWORK QUEUES ?
         SUB (7036)
         SPA
         JMP I 16SPCERR
         SUB (7055-7036)
         SZM
         JMP I 16SPCERR
FIXHD2,  JDA TRACK
         030000
         JSP WAIT4      /GET USER CORE BACK.
         LAC (210000)
         SZF 1
         LAC (270000)
         DIP FIXCMD+1
         LAC I IO
         DAP FIXCMD+1
         LAW 7777
         AND I AC
         IOR SCORE
         DAC FIXCMD+2
         JSP GO
         I+I+I FIXCMD
         JMP SWAPE1

FIXCMD,  560001
         0
         0
         100063
         0

```

/RELEASE HALF TRACK

```

RTRACK:  JSP ,FSET
          STF 2
          JSP WHERE
          JDA TRACK
          000000
          LAC HTR
          CLI"U"CMI      /MARK BITS AS NOT IN CORE
          SAD TRACKA
          DIO TRACKA
          SAD TRACKB
          DIO TRACKB
          SAD TRACKC
          DIO TRACKC
          SAD TRACKD
          DIO TRACKD
          CLF 6
          RAR 89
          AND (6000)
          DAP RTRKC0+1
          DAP RTRKC2+1    /SET STARTING DRUM ADDRESSES
          ADD (262000)
          DAC RTRKE
RTRK1,   DZM RTRKC0+7    /SET CORRECT HALT
          JSP WAIT2
          30000 RTRKC0
          170000 RTRKC1+7
          STF 6
          LAW 100
          ADD RTRKC0+1    /INCREMENT TO NEXT HEAD
          DAP RTRKC1+1
          DZM RTRKC1+7
          JSP WAIT2
          30000 RTRKC1
          JDA RTRKC0+7
          LAW 100
          ADD RTRKC1+1
          DAP RTRKC0+1
          SAS RTRKE      /WAS THIS THE LAST HALF TRACK
          JMP RTRK1      /NO
          LAW BUFF      /YES BUILD A BIT TABLE
          DAP RTRK2+1
RTRK2,   LIO THREE
RTRK2+1, DIO .
          IDX RTRK2+1
          SAS (DIO BUFF 76
          JMP RTRK2+1
          JSP WAIT2
          30000 RTRKC2
          170000 RTRKC1+7
          SZF I 3
          DZM I HTR
          JSP WAIT4
          JMP SWAPE1

```


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RTRKC0, 560001
RTRKC0+1, 260000
630000 ZERO

050001
MZERO, 777777
046276
100001

RTRKC0+7, 000000

RTRKC1, 560001
RTRKC1+1, 260000
630000 ZERO

050001
777777
046276
100001

RTRKC1+7, 000000

RTRKC2, 560001
RTRKC2+1, 260000
630000 RTRKF

040002
630000 BUFF
040076
630000 (=1776
100001
000000

RTRKF, 400003
400003

RTRKE, 0

/END CONDITION STORAGE

```

.FSET,   DAP .FSETX
          JSP .SETUP
          LAI
          AND (6
          RAR 38
          LIA
          SAS (600000
.FSETX,   JMP .
          JMP I 16SPCERR

```

```

/LINK SET RETURN 3
.SETUP,   DAC .SETX
          LAC I RSTAT
          DAP 16STAT
          ADD MSTAT
          DAC 16USER
          LAC I (DCH 71)
          DAC RPC
          IDA
          SZF 6
          IDA
          CLL"U"SZL
          IDA
          DAC I (DCH 71)
          JMP I .SETX
.SETX,    0
RPC,      0

```

```

/ROUTINE TO CALCULATE IOPMAX FOR READS
MCHECK,   0 /ENTER WITH COUNT IN AC
          DAP MCHX
          LAC MCHECK
          LIO ONE
          ADD CORADR
          SUB MAXM
          SPQ
          JMP MCH1
          SAS ONE
          SWP"U"STF 1
MCH1,     ADD MAXM
MCHX,     JMP .

```

/READ INFORMATION
 /FLAG 1=1,BLOCK,0,ITEM
 /FLAG 2=1 PUT ONLY DRAS IN USER CORE

```

RDINF:   JSP ,FSET           /INITIALIZE IOT
         LAW 7777           /CHECK IF BELOW BOUND
         AND I (DCH 70)
         SUB BOUND
         SPA
         JMP I 16SPCERR
         LAC I (DCH 72) /CHECK IF LEGAL DRUM ADDRESS
         SPA
         JMP I 16SPCERR
         SAR 1S
         IAI
         JDA TRACK
ZERO,    000000           /MOVE BOOM TO PROPER LOCATION
RDINF3,  JSP WAIT4
         LIO I IO
         DIO DRMPTR
         LAW 7777
         AND I AC
         DAP RAITEM
RDINF1,  IOR SCORE
         DAC CORADR
         DAP CORPTR
         SUB C61           /SET UP DAC COMANDS IN ITEM LOOP
         DAC RAIDAC
         LAW 7777
         AND I IOPMAX
         SZA I
         LAW 7777
         IOR SCORE
         DAC MAXM
         DAC COUNT       /EXTRA SET FOR READ ITEM CASE
         SUB CORADR
         SPA
         JMP MAXERR       /IOPMAX<STARTING LOCATION
         SZF I 1
         JMP RIT          /READ ITEM
         CLF 7
         LAC DRMPTR       /SET HEAD=SECTOR
         DAP RBKC+1
         LAW 62           /CALCULATE WORD COUNT WANTED
         JDA MCHECK
         SUB CORADR
         DAP RBKC+5       /SET WORD COUNT
         JSP GO
         I I I RBKC
         LAC PTR1
         DAC I IO
         SAS DRMPTR
         JSP ILLITM
         JMP RIT4

```

RBKC,	560001
RBKC+1,	200000+.
	630000 PTR1
C40001,	040001
CORADR,	000000
	100000 .
	000000

RIT,	LAW ILLITM	/MARK IN COMPARE MODE FOR ABNORMAL ROUTINE
	DAP COMPM	
	LAW 61	
	DAP RITC1+7	
	CLC"U"CLF 3	
	CLF 5	
	DAC RITC1+4	
	DAC RITC1+6	
	LAC (040001)	
	DAC RITC1+5	
	DIP RAITEM	
	DZM I ERCD2	
	LAC (30000)	
	ADD MAXM	
	SZF 2	
	DAC MAXM	
	LAC DRMPTR	
	DAC PTR1	
	DAC PTR2	
	DAC LTRK	
RIT1,	LAC PTR1	
	SZF I 5	
	SAD ONE	/WAS THIS LAST BLOCK
	JMP RTRN	/YES
	SPQ	
	JSP ILLITM	/ILLEGAL ITEM
	DAP RITC1+1	
	XOR LTRK	/IS TRACK THE SAME
	AND (770000)	
	SZA I	
	JMP RITS	/CHANGING TRACKS?
	LAC PTR1	/YES
	JDA ,READ	
RITS,	LAW 61	
	ADD RAIDAC	
	DAC RAIDAC	
	SZF I 2	
	DAC RITC1+6	
	ADD (60)	
	SUB COUNT	
	SPA	
	JMP RIT7	
	SUB C61	
	CMA"U"STF 5	
	DAP RITC1+7	
RIT7,	LAC PTR1	
	SZF 2	
	DAC I RAITEM	
	IDX RAITEM	
	IDX I ERCD2	
	JSP GO	
	I I I RITC1	

```

SZF 3
JMP RIT1
STF 3
LAC CPTR2      /RESET D.C. PROGRAMS
DAC RITC1+4
LAC CXOR
DIP RITC1+5
LAC PTR1       /CHECK IF START OF ITEM
SMA
JSP ILLITM     /ILLEGAL ITEM
XOR B0
SAD PTR2       /IS IT A DOG-IN-THE-MANGER?
JSP ILLITM
DAC PTR1
LAC I CORPTR   /GET COUNT
SZF 2
LAW 7742
JDA MCHECK
SUB ONE
DAC COUNT
SUB (60)       /ARE WE DONE
SUB RAIDAC
SZM
JMP RIT1

RIT4,          STF 5
RTRN,          SZF 1
               JMP MAXERR
               SZF I 2
               SZF 5
               JMP SWAPE1
ILLITM,        LAW 1000
               JDA 1ERR

.READ,         0
               DAP .READX
               LAC .READ
               DAC LTRK
               RAL 8S
               AND (177
               ADD THIRD
               JDA READ
.READX,        JMP .

RITC1,         560101
               200000+.
               630000 PTR1
RITC1+3,       040001
               0
               0
RITC1+6,       0
               100000 .
               000000

RAITEM,        IOR .
RAIDAC,        0

```

/XOR 1 OR IOR 1

/EXPUNGE INFORMATION
 /FLAG 1=1_BLOCK,0_ITEM
 /FLAG 2=1_NEGLECT OWN=WRDS(MUST BE HELD)

EPINF:	JSP .FSET	
	JSP WHERE	/BUILD UP POINTER
	JDA TRACK	
	020000	
	JSP WAIT4	
	LAC I AC	
	DAP CORPTR	
	LAC I IO	
	DAC DRMPTR	
EPINF1,	DAP EXB0+1	/SECOND ENTRY FOR PARAMETER EXPUNGE,ETC.
	DAP EXB1+1	
	JSP GO	/READ BLOCK INTO BUFF
	I I I EXB0	
	SZF I 1	
	JMP EXPITM	/EXPUNGE ITEM
	LAC BUFF	
	SAS DRMPTR	
	JSP ILLITM	
	JSP GO	
	I I I EXB1	
	LAC DRMPTR	/PUT BIT BACK IN TABLE
	JDA EXP12	
	JSP WAIT	/WAIT FOR FINISH OF BLOCK ERASE
	JMP SWAPE1	
RWITC2,		
EXB0,	560001	
	200000+.	
	630000 BUFF	
	100063	
	000000	
EXB1,	560001	
	260000+.	
	630000 ZERO	
C50001,	050001	
	777777	
	100062	
	000000	

```

WHERE,      DAP  WHERX
             LAC  I (DCH 72)
             SPA
             JMP  I 16SPCERR
             SAR  1S
             IAI
             DAC  WHER3
             RAL  9S
             AND  C777
             ADD  (FRELST)
             DAC  WHRPTR
             LAC  I WHRPTR
             RAL  1S
             SPA
             JMP  WHER1
             SZF  2           /FREE=DO WE WANT TO NEGLECT OWNWRDS
             JMP  I NOTYOURS /YES=ILLEGAL
WHER2,      LAC  WHER3
WHERX,      JMP  .
WHER1,      LAC  I WHRPTR
             XOR  B1
             AND  (340177)
             SAD  16USER
             JMP  WHER2
             XOR  (DCH 200)
             SZF  I 2
             SAS  I NETHND
             JMP  I NOTYOURS
             JMP  WHER2           /NETHANDLER QTRACKS ARE ACCESSIBLE
WHRPTR,    0
WHER3,      0

EXPITM,     JSP  CKITM
EXPT1,      LAW  ILLITM
             DAP  COMPM
             CLC           /DON'T COMPARE FIRST BLOCK
             DAC  EXPC0 4
             LAC  (040001)
             DAC  EXPC0 5
             LAC  DRMPTR
             DAC  PTR1      /SET UP DRA FOR COMPARING
             SZF  I 3       /DOG IN THE MANGER REQUESTED?
             JMP  EXP5      /NO
             IOR  B0
             DAC  FEXP
             DAP  EXPC4+1
             JSP  GO
             I I I EXPC4
             LAC  BUFF
             AND  (377777)
             DAC  DRMPTR
EXP4,       LAC  (JSP I PTR1) /WE SHOULD COMPARE NOW
             DAC  EXPC0 4
             LAC  (060001)
             DAC  EXPC0 5

```



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EXP5,    LAC DRMPTR
          SAD ONE
          JMP APREXT
          DAC LTRK
          RAL 8S
          AND (177
          ADD THIRD
          JDA REWRITE

EXP2,    LAW BUFF          /SET POINTER TO FIRST OF BUFFER
          DAP EXP3
          CLF 7

EXP9,    LAC DRMPTR
          AND (377777)
          DAC DRMPTR
          SPQ
          JSP ILLITM
          DAP EXPC0+1

EXP.11,  SAD ONE
          JMP EXP6          /THE END

EXP3,    DAC .             /O.K.-STORE IN BUFFER
          XOR LTRK          /DOES IT CHANGE QUARTER TRACK
          AND (376000)
          SZA
          JMP EXP6          /YES
          IDX EXP3
          SAD (DAC BUFF 477)
          JMP EXP7
          JSP GO
          I I I EXPC0
          LAC (060001)
          DAC EXPC0 5
          LAC (JSP I PTR1)
          DAC EXPC0 4
          JMP EXP9

EXP7,    LAW EXP2
          JMP EXP1

```

/PUT BLOCKS BACK IN TABLE
 /ASSUME ON RIGHT TRACK WITH BITS IN CORE
 /BLOCKS IN BUFF
 /TERMINATE BY -0

EXP6, LAW EXP5
 EXP1, DAP EXP1X
 LAW BUFF
 DAP EXP11
 CLC"U"CLF 6
 XCT EXP3

EXP11, LAC , /GET A BLOCK ADDRESS

SPA

EXP1X,

JMP ,

DAP EXPC2+1

/PLACE HEAD AND SECTOR

JSP GO

I I I EXPC2

XCT EXP11

JDA EXP12

/RESET THE BIT TABLE

IDX EXP11

JSP WAIT

JMP EXP11

/CHECK REWRITE NUMBER, OWNWORD ETC

```

CKITM,    DAP CKITMX
          LAC BUFF      /IS IT AN ITEM
          SMA
          JSP ILLITM
CXOR,     XOR B0
          SAD DRMPTR
          JSP ILLITM
          SZF 2
          JMP CKITMX
          LAC BUFF+1
          SAS I OWNWRD
          JMP RTERR
          IDX CORPTR
          LAC I CORPTR
          SAD BUFF+3
CKITMX,   JMP .
RTERR,    LAW 2000
          JDA 1ERR

```

/PUT THINGS BACK ON BIT TABLE

```

EXP12,    0
          DAP EXP12X      /AC=17 BIT DRUM POINTER
          LAW 77
          AND EXP12
          ADD BITPTR      /GET POINTER TO WORDS THIS SECTOR
          DAC EXP121
          LAW 1700
          AND EXP12
          SAR 6S          /GET HEAD
          ADD (B0
          DAP EXP124      /GET POINTER TO RIGHT BIT
          LAC I EXP121
EXP124,   XOR .
          DAC I EXP121
          LAC (757777)
          AND I ONTRACK
          DAC I ONTRACK
          LAW I 1          /REST COUNT OF BLOCKS
          ADD I CNTPTR
          DAC I CNTPTR
          SAD (-1776)
          DZM I ONTRACK
EXP12X,   JMP .
EXP121,   0

```

B0, 400000
 B1, 200000
 B2, 100000
 B3, 40000
 B4, 20000
 B5, 10000
 B6, 4000
 B7, 2000
 B8, 1000
 C400, 400
 C200, 200
 C100, 100
 C40, 40
 C20, 20
 C10, 10
 FOUR, 4
 TWO, 2
 ONE, 1
 C60001, 60001

EXPC0, 560101
 200000+
 630000 DRMPTR

EXPC0+3, 040001
 0

/IOR 1 OR XOR 1

EXPC0+6, 777777
 100061
 000000

EXPC2, 560001
 EXPC2+1, 260000+
 630000 ZERO
 050001

777777
 100062
 EXPC2+6, 000000

EXPC4, 560001
 260000+
 630000 FEXP
 040001
 777777
 100062
 000000

FEXP, 0

```

/WRITE NON-ADDRESSED INFORMATION
/FLAG 1=1 BLOCK, 0 ITEM
/FLAG 2=1 HELD, 0 NOT HELD
/FLAG 3=1 AND FLAG 1=1 WRITE A ZEROED BLOCK
/FLAG 3=1 AND FLAG 1=0 WRITE OUT THE BIT TABLE (RESET THE FRELST)
/ AND THEN WRITE THE NON-ADDRESSED ITEM

```

```

WNAI1:    JSP .FSET
          DIO TRACK
          LAC B5
          SZF 2
          LAC B3
          DIP WNAI1
          LAW 1          /BLOCK OR ITEM
          SZF 1
          JMP WNAI2
          LAW 7777       /ITEM=GET WORD COUNT
          AND I (DCH 70)
          DAC WNAI4
          LAW I 1
          ADD I WNAI4
          ADD WNAI4
          AND (770000)
          SZA
          JMP I 16SPCERR
          LAC I WNAI4
          SAS ONE
          SPQ
          JMP I 16SPCERR
          SCR 98
          SCR 88
          DIV C61
C61,      61
          SNI I
          IDA
WNAI2,    IOR TRACK
          JDA TRACK
WNAI1,    0
WNAI3,    JSP WAIT4     /WAIT FOR CORE
          LAW 7777
          AND I AC
          IOR SCORE
          DAC WITC0+6
          DAC WNBC+4
          SZF I 1
          JMP WNAI6      /ITEM
          CLC
          SZF 3          /DO WE WANT TO WRITE ZEROS
          DAC WNBC+4     /YES
          LAC BUFF
          DAP WNBC+1     /SET HEAD AND SECTOR
          DAC I IO
          JSP GO         /GO!
          I I I WNBC
          JMP SWAPE1
WNAI4,    0

```

WNBC, 560001
 260000+.
 630000 BUFF
 040001
 WNBC+4, 0
 100062
 000000

WNAI6,	SZF I 3	/WANT TO WRITE OUT BIT TABLES ?
	JMP WIT	/NO. NORMAL CASE.
	LAC THIRD	
	SUB (FRELST)	
	SAR 7S	
	LIA	
	ADD (TRACKA)	
	DAP .+1	
	LAC .	
	SAS ONTRACK	/ALL HEALTHY ?
	JMP WIT	/STRANGE.
	JSP BTWRIT	/SET UP COMMANDS
	JMP WIT	/ STRANGE.
	JSP GO	/WRITE THEM OUT
	I I I NDONC1	
	JMP WIT	/ON TO NORMAL WRITE NON-ADR ITEM

```

WIT,      CLL"U"CML      /FIRST BLOCK OF ITEM
          LAC I OWNWRD
          DAC PTR2
          LAC BUFF      /SAVE DRA FOR BACK PTRS
          DAC PTR3
WIP,      LAW BUFF
          DAP WIP12
          XCT WIP12
          SAD MZERO
          JSP WIP1      /GET MORE BLOCKS
          DAC PTR1      /SET UP FORWARD PTR
          DAC LTRK
          IDX WIP12
          CLF 7
WIP2,     LAC PTR1
          DAP WITC0+1
          XOR LTRK
          AND C376000
          SZA
WIP12,    JSP WIP3
          LAC .
          SPA
          JSP WIP1
          SZL
          IOR B0      /FIX UP FIRST TAG
          DAC PTR1
          IDX WIP12
          JSP GO
          I I I WITC0
          LAC PTR3
          CLL"U"SZL
          DAC PTR2
          SZF 5
          JMP WIP6
          LAW 61
          ADD WITC0+6
          DAP WITC0+6
          JMP WIP2

```

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WITC0, 560001
WITC0+1, 260000+.
630000 PTR1
040001
CPTR2, 630000 PTR2
040001
WITC0+6, 0
100061
WITC0+10, 000000


```

/ MOVE BOOM FOR WRITE
WIP3,      DAP WIP3X
           JSP WAIT
           LAC PTR1
WIP5,      DAC LTRK
           RAL 8S
           AND (177
           ADD THIRD
           JDA REWRITE
WIP3X1,    LAC PTR1
WIP3X,     JMP .

```

```

/ GET MORE BLOCKS FOR WRITE

```

```

WIP1,      DAP WIP3X
WIP11,     LAC FBLOCK      / ARE WE DONE
           SMA
           JMP WIP4        / DONE
           LAC PTR1
           JDA GBLOCK      / GET MORE BLOCKS
           LAW BUFF
           DAP WIP12
           XCT WIP12
           LIO PTR1        / SET UP SO MOVES BACK TO OLD TRACK BUT SEES NEW POI
           DAC PTR1
           LAI
           JMP WIP5

```

```

WIP4,      LAW 1          / TO WRITE LAST BLOCKS
           DAC PTR1
           STF 5
           JMP WIP3X1

```

```

WIP6,      LAC PTR3      / SET USER IO
           DAC I IO
APREXT,    JMP SWAPE1

```

```

SPPTR1, PTR1,           0
SPPTR2, PTR2,           0
PULPTR, PTR3,           0
DRMPTR, 0
CORPTR,  IOR .
MAXM,    0
COUNT,  0
LTRK,    0
APRFLG,  0
PATFLG,  0

```

```

/REWRITE INFO
/FLAG 1=1 BLOCK, 0 ITEM
/FLAG 2=1 DO NOT CHECK REWRITE NO. (MUST BE HELD)
/FLAG 3=1 APR (IF ITEM)

```

```

RWINF:  JSP .FSET      /INITIALIZE AND DECODE THIRD
        JSP WHERE     /FIGURE OUT WHERE DRUM POINTER IS
        JDA TRACK     /MOVE BOOM IN A READ FASHION
        000000
        JSP WAIT4      /WAIT FOR CORE TO SHOW UP
        LAW 7777
        AND I AC
        IOR SCORE
        DAC RWTC0+4
        DAC RWITC0+6   /SET UP DAC COMMANDS
        DAP CORPTR
        LAC I IO
        DAC DRMPTR
        DAP RWTC0+1    /SET UP HEAD AND SECTOR
        DAC PTR1
        DAC PTR3
        DAP RWITC2+1
        AND (1777)
        SAS ONE
        SZA I
        JMP ILLITM
        SZF I 1        /ITEM OR BLOCK
        JMP RWIT       /ITEM
        LAW ILLITM     /BLOCK=SET FOR COMPARE MODE
        DAP COMPTX
        LAC (060001)   /ARE WE WRITING OFF HELD OR FREE
        LIO I ONTRACK
        RIL 1S
        SPI
        LAC (040001)
        DAC RWTC0+3
        CLC
        SZF 3
        DAC RWTC0+4
        JSP GO
        I I I RWTC0
        LAC PTR1
        DAC I IO
        JMP SWAPE1

RWTC0,  560101
        220000+
        630000 PTR1
        0              /IOR 1 OR XOR 1

RWTC0+4, 0
        100062
        000000

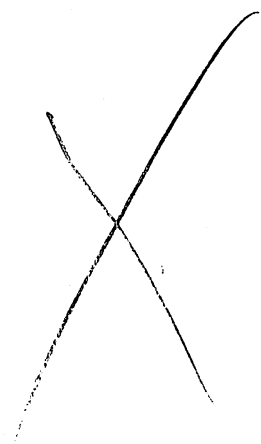
```

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/DATA CHANNEL COMMANDS FOR REWRITE ITEM

RWITC0, 560001
220000+,
630000 PTR1
040001
630000 PTR2
040001

RWITC0+6, 0
100061
000000



/REWRITE ITEM
/FLAG 3=1, APR ITEM

RWIT,	LAW 7777	/DOES ITEM GO OFF THE END OF CORE?
	AND CORPTR	
	ADD I CORPTR	
	SUB ONE	
	AND (770000)	
	SZA	
	JSP ILLSPC	/LENGTH TOO LONG
	LAC I CORPTR	/IS IT A LEGAL BLOCK COUNT
	SAD ONE	
	JSP ILLSPC	/ILLEGAL ITEM LENGTH
	MUL ONE	/PUT IN DIVIDE FORMAT
	DIV C61	
C62,	62	
	SNI I	
	IDA	
	SPQ	/IS IT A LEGAL NUMBER
	JSP ILLSPC	/NO
	CMA	
	DAC FBLOCK	
	SZF 2	
	JMP ARWIT	
	JSP GO	/GET BLOCK INTO BUFF=BUFF 62
	I I I RWITC2	
	JSP CKITM	
	IDX I CORPTR	/UPDATE THE REWRITE NUMBER
ARWIT,	CLL"U"CML"U"SCF	
	LAC I OWNWRD	
	DAC PTR2	

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FBI/DOJ

```

RWIT1,  LAC PTR1      /SET HEAD AND SECTOR
        AND (377777
        DAC PTR1
        SAD ONE      /IS THIS LAST BLOCK OF PREVIOUS ITEM
        JMP RWIT2     /YES-GO WRITE MORE
        DAP RWITC0 1
        SZF 1
        CLL
        LIO PTR3
        SZL I
        DIO PTR2
        LAW 61
        ADD RWITC0 6
        SZF 1
        DAP RWITC0 6
        STF 1
        LAC PTR1
        XOR LTRK     /DO WE WANT TO CHANGE TRACKS
        AND (376000
        SZA
        JSP WIP3      /YES-GO MOVE THE BOOM
        JSP GO
        I I I RWITC0
        ISP FBLOCK    /ARE WE DONE WITH BLOCKS WANTED
        JMP RWIT1
        LIO RWITC0+6
        LAC RWITC0+1
        DAP WITC0+1   /REWRITE LAST BLOCK WITH PROPER TAG
        DIO WITC0+6
        LAC PTR1
        AND (377777
        SAD ONE
        JMP APREXT    /TAG ALREADY SET PROPERLY
        DAC DRMPTR
        LAW 1
        SZL
        IOR B0
        DAC PTR1
        JSP GO
        30000 WITC0
        LAC (JSP I PTR1) /MAKE COMPARE STUFF WORK
        DAC EXPC0 4
        LAC (060001)
        DAC EXPC0 5
        LAC PTR3
        DAC PTR1
        LAW EXP2
        JMP WIP3      /GO GET RID OF THE REST OF THE BLOCKS
RWIT4,

```

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```
/WRITE MORE BLOCKS=FIRST HALF
RWIT2,    LIO RWITC0+6    /GET DAC
          LAC RWITC0+1
RWIT7,    DAP BUFF
          LAC LTRK
          DIP BUFF
          CLC
          DAC BUFF 1
          DIO WITC0+6
          JMP WIP          /GO WRITE AS A WRITE ITEM
```

```
/INDEX IOT (SEE CORE 15)
INDEX2,   JSP ,FSET
          SZF I 4
          SZF 5
          JMP I NOTYOURS
          LAI
          JDA TRACK
          000000
          JSP WAIT4
          JMP I INDEX1
```

START XX=JMP